

Brute force Investigation with Splunk

Analysis of logs investigating brute force attack in Splunk

Aayush Dhakal

Bachelors in Computer Networking and IT Security

Islington College, Nepal

August 2026

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1 Scenario

The Blue Team lab from BTLO provided a log file for investigation on an attempted RDP brute force. A large number of Audit Failure events in the Windows Security Event log attack was identified.

2 Analysis

The CVS file (BTLO_Bruteforce_Challenge.csv) of the logs was uploaded to Splunk.

The brute force attacks in Windows Security Event Log can be identified with event ID 4624 and 4625.

← Windows Security Log Event ID 4624 →

4624: An account was successfully logged on

On this page

- Description of this event
- [Field level details](#)
- [Examples](#)

This is a highly valuable event since it documents each and every successful attempt to logon to the local computer regardless of logon type, location of the user or type of account. You can tie this event to logoff events [4634](#) and [4647](#) using Logon ID.

Win2012 adds the Impersonation Level field as shown in the example.

Win2016/10 add further fields explained below.

Operating Systems	Windows 2008 R2 and 7 Windows 2012 R2 and 8.1 Windows 2016 and 10 Windows Server 2019 and 2022 Windows Server 2025
Category • Subcategory	Logon/Logoff • Logon
Type	Success
Corresponding events in Windows 2003 and before	528 , 540

← Windows Security Log Event ID 4625 →

4625: An account failed to log on

On this page

- Description of this event
- [Field level details](#)
- [Examples](#)

This is a useful event because it documents each and every failed attempt to logon to the local computer regardless of logon type, location of the user or type of account.

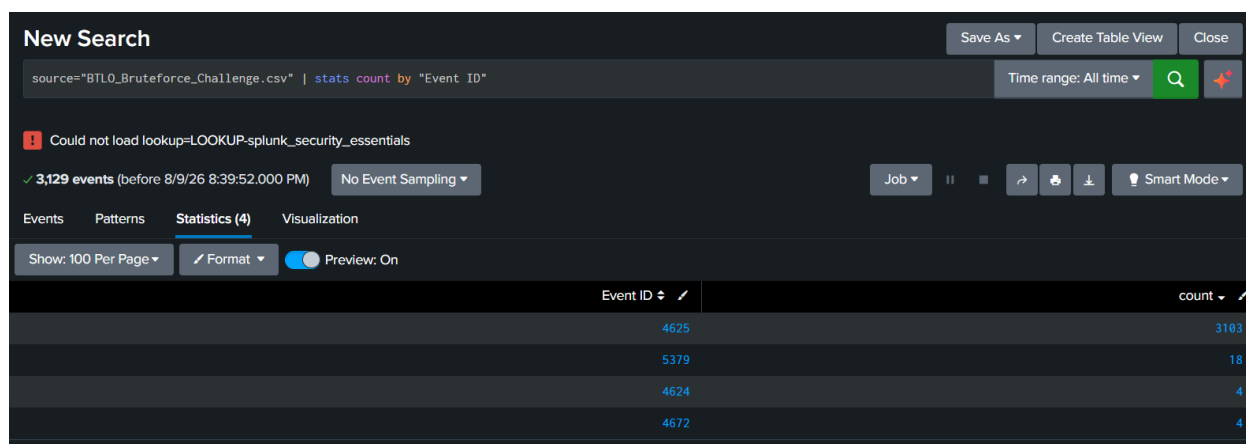
Operating Systems	Windows 2008 R2 and 7 Windows 2012 R2 and 8.1 Windows 2016 and 10 Windows Server 2019 and 2022 Windows Server 2025
Category • Subcategory	Logon/Logoff • Logon
Type	Failure
Corresponding events in Windows 2003 and before	529 , 530 , 531 , 532 , 533 , 534 , 535 , 536 , 537 , 539

Figure 1 Event ID 4624 and 4625

Following query was used for this investigation:

```
source="BTLO_Bruteforce_Challenge.csv" | stats count by "Event ID"
```

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The screenshot shows the Splunk 'New Search' interface. The search bar contains the query: `source="BTLO_Bruteforce_Challenge.csv" | stats count by "Event ID"`. The results are displayed in a table with two columns: 'Event ID' and 'count'. The table shows four rows of data.

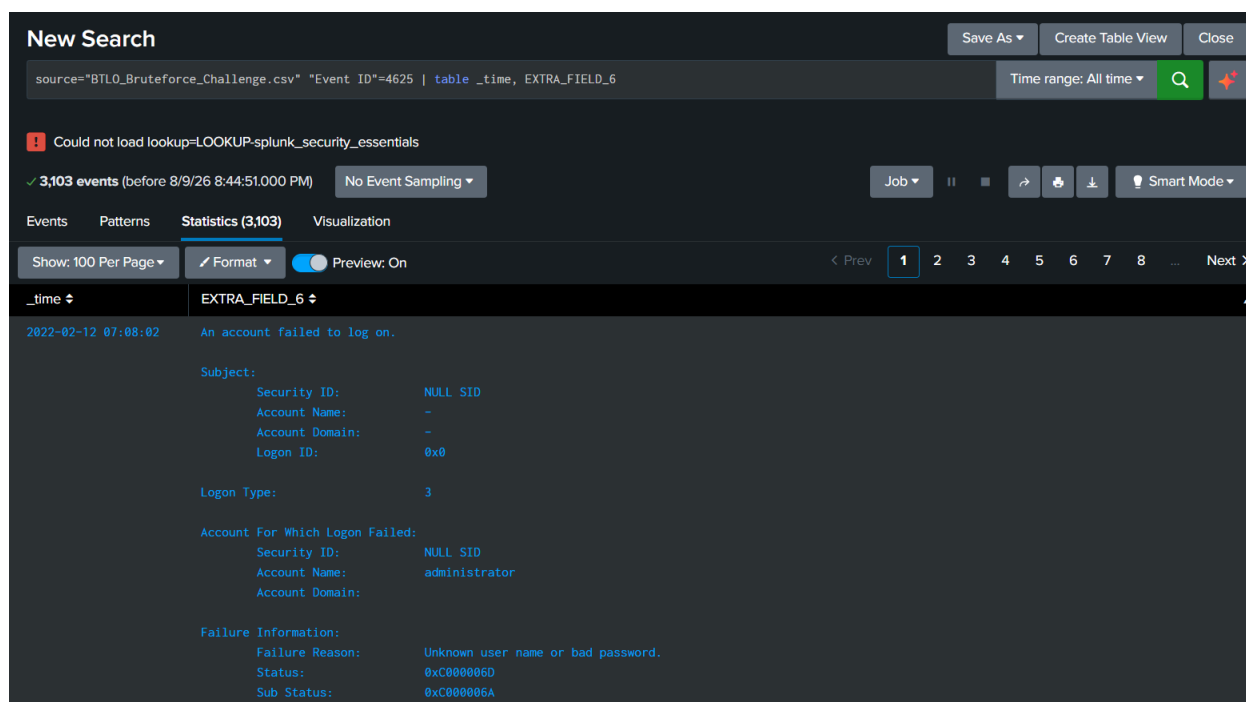
Event ID	count
4625	3103
5379	18
4624	4
4672	4

Figure 2 Query I result

There was a very high number of unsuccessful logons (event id 4625). Four successful logons were also made (event id 4624).

The logs had a field called EXTRA_FIELD_6. The following query was used to investigate this field of the failed logons:

```
source="BTLO_Bruteforce_Challenge.csv" "Event ID"=4625 | table _time, EXTRA_FIELD_6
```



The screenshot shows the Splunk 'New Search' interface with the query: `source="BTLO_Bruteforce_Challenge.csv" "Event ID"=4625 | table _time, EXTRA_FIELD_6`. The results are displayed in a table with two columns: '_time' and 'EXTRA_FIELD_6'. The table shows a single row of data.

_time	EXTRA_FIELD_6
2022-02-12 07:08:02	An account failed to log on. Subject: Security ID: NULL SID Account Name: - Account Domain: - Logon ID: 0x0 Logon Type: 3 Account For Which Logon Failed: Security ID: NULL SID Account Name: administrator Account Domain: Failure Information: Failure Reason: Unknown user name or bad password. Status: 0xC000006D Sub Status: 0xC000006A

Figure 3 Query II result

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The field had information like Account Name, Failure Reason, Source Network Address and Source Port. Following query was used to investigate further:

```
source="BTLO_Bruteforce_Challenge.csv" "Event ID"=4625

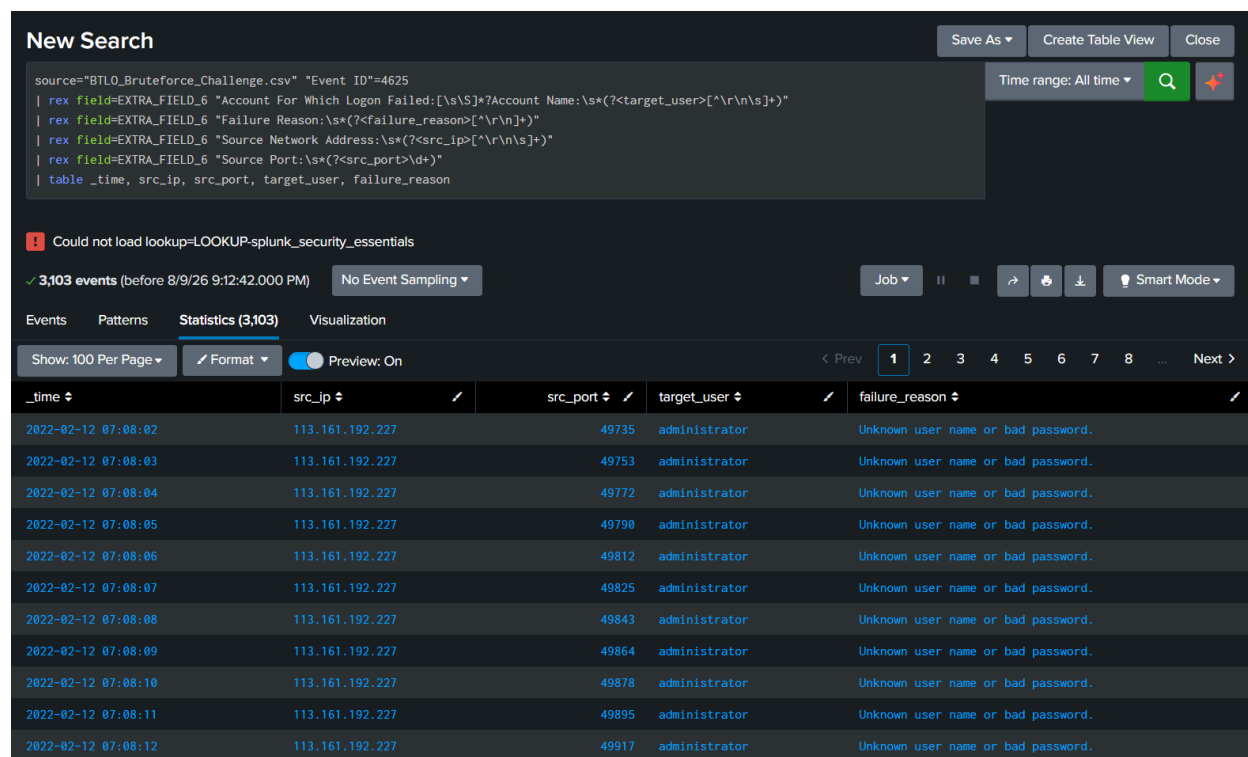
| rex field=EXTRA_FIELD_6 "Account For Which Logon Failed:[\s\S]*?Account
Name:\s*(?<target_user>[^\r\n\s]+)"

| rex field=EXTRA_FIELD_6 "Failure Reason:\s*(?<failure_reason>[^\r\n]+)"

| rex field=EXTRA_FIELD_6 "Source Network Address:\s*(?<src_ip>[^\r\n\s]+)"

| rex field=EXTRA_FIELD_6 "Source Port:\s*(?<src_port>\d+)"

| table _time, src_ip, src_port, target_user, failure_reason
```



The screenshot shows the Splunk interface with a search query and its results. The query is: `source="BTLO_Bruteforce_Challenge.csv" "Event ID"=4625 | rex field=EXTRA_FIELD_6 "Account For Which Logon Failed:[\s\S]*?Account Name:\s*(?<target_user>[^\r\n\s]+)" | rex field=EXTRA_FIELD_6 "Failure Reason:\s*(?<failure_reason>[^\r\n]+)" | rex field=EXTRA_FIELD_6 "Source Network Address:\s*(?<src_ip>[^\r\n\s]+)" | rex field=EXTRA_FIELD_6 "Source Port:\s*(?<src_port>\d+)" | table _time, src_ip, src_port, target_user, failure_reason`. The results table shows 12 events, all from 2022-02-12 07:08:02 to 07:08:12, with source IP 113.161.192.227, source port 49735 to 49917, target user 'administrator', and failure reason 'Unknown user name or bad password.'

_time	src_ip	src_port	target_user	failure_reason
2022-02-12 07:08:02	113.161.192.227	49735	administrator	Unknown user name or bad password.
2022-02-12 07:08:03	113.161.192.227	49753	administrator	Unknown user name or bad password.
2022-02-12 07:08:04	113.161.192.227	49772	administrator	Unknown user name or bad password.
2022-02-12 07:08:05	113.161.192.227	49790	administrator	Unknown user name or bad password.
2022-02-12 07:08:06	113.161.192.227	49812	administrator	Unknown user name or bad password.
2022-02-12 07:08:07	113.161.192.227	49825	administrator	Unknown user name or bad password.
2022-02-12 07:08:08	113.161.192.227	49843	administrator	Unknown user name or bad password.
2022-02-12 07:08:09	113.161.192.227	49864	administrator	Unknown user name or bad password.
2022-02-12 07:08:10	113.161.192.227	49878	administrator	Unknown user name or bad password.
2022-02-12 07:08:11	113.161.192.227	49895	administrator	Unknown user name or bad password.
2022-02-12 07:08:12	113.161.192.227	49917	administrator	Unknown user name or bad password.

Figure 4 Query III result

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The failed login attempt was made on administrator account. The reason for failure was unknown username or bad password, which indicates brute force attack. The source network address was 113.161.192.227. According to VirusTotal, the IP address belongs to Vietnam.

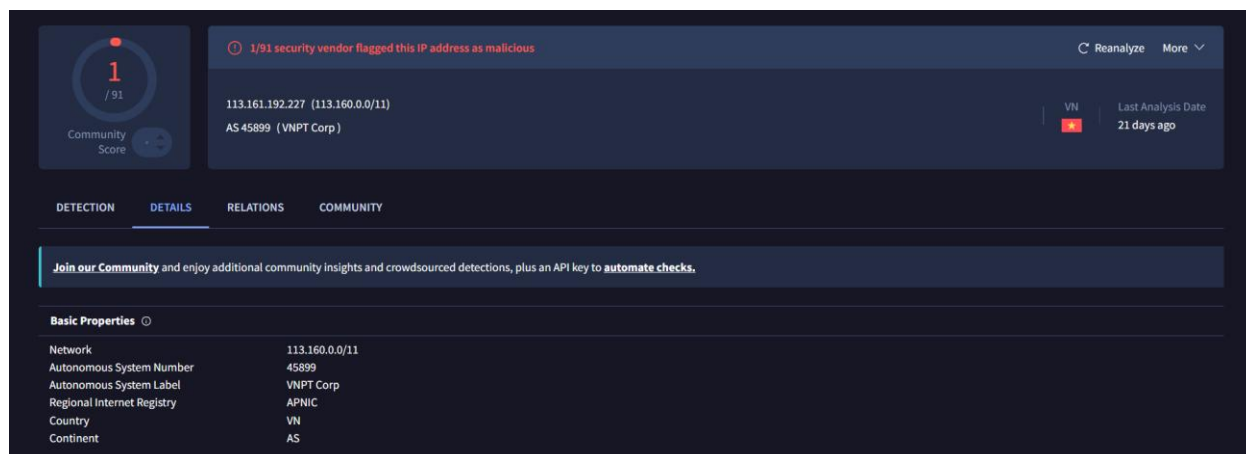


Figure 5 VirusTotal review of IP address

There were different source ports in each log. Following query was used to list the ports. Sorting by ascending and descending gave the smallest and biggest port number.

```
source="BTLO_BruteForce_Challenge.csv" "Event ID"=4625  
| rex field=EXTRA_FIELD_6 "Source Port:\s*(?<src_port>\d+)"  
| stats count by src_port
```

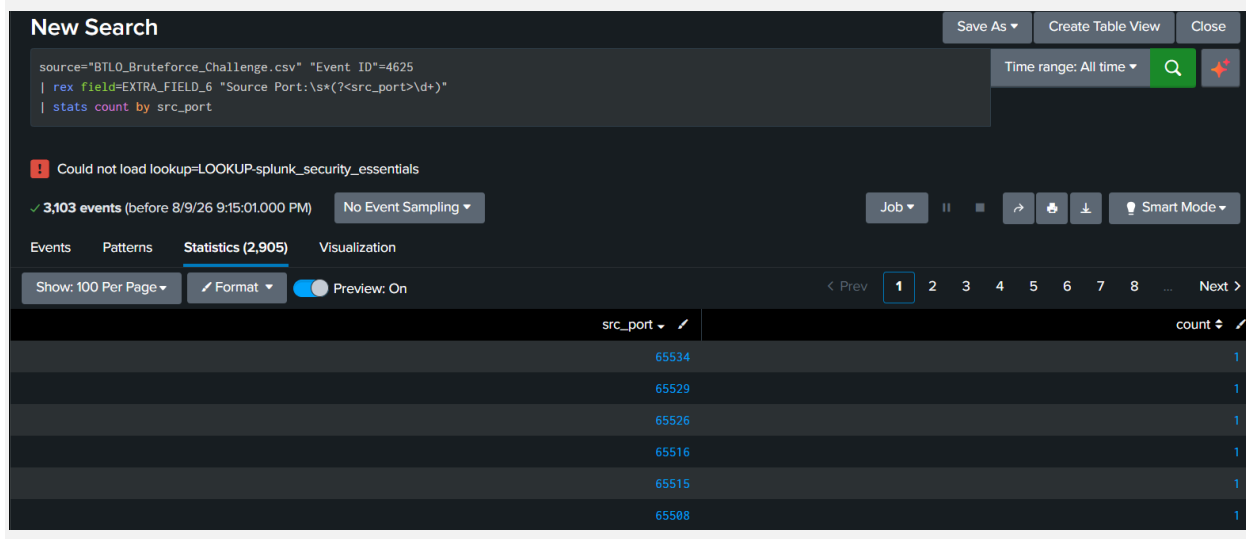


Figure 6 Query V result

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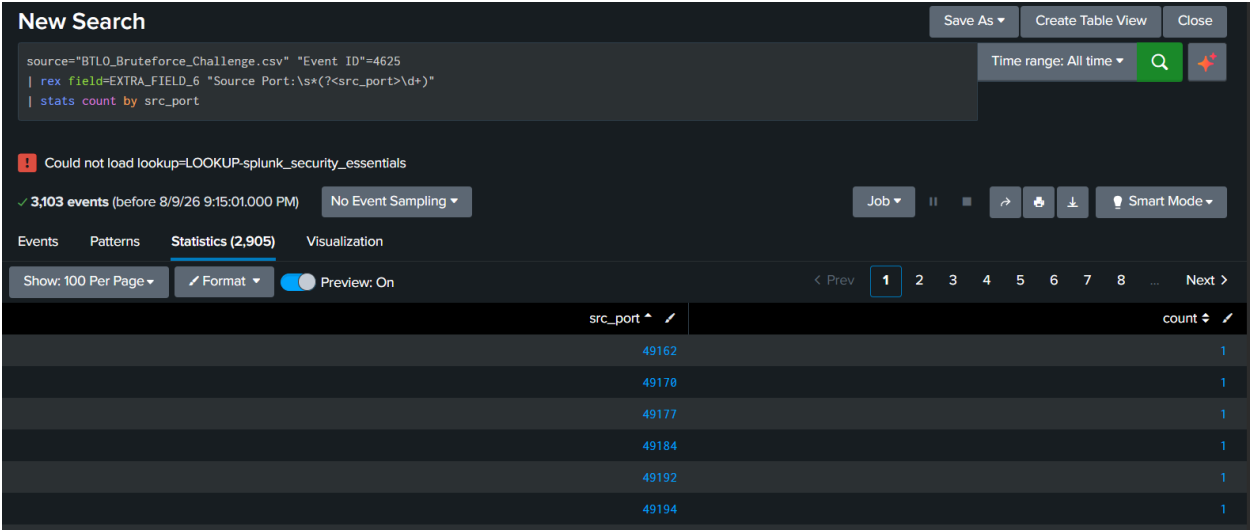


Figure 7 Ascending order

The range of source port was 49162 to 65534.

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3 Findings

1. 3,103 unsuccessful login attempts were made by the attacker.
2. The attacker was using IP address 113.161.192.227 which originated from Vietnam.
3. Various ports ranging from 49162 to 65534 were used by the attacker to brute force.